

DAY-4

TASK 2: VIRTUAL TRY-ON AI MODELS:

Virtual try-on systems use Artificial Intelligence and Computer Vision techniques to allow users to digitally visualize clothing on their bodies before purchasing. These systems combine pose detection, body segmentation, cloth warping, and image generation to create realistic clothing simulations. AI models in virtual try-on applications help improve online shopping experiences by reducing size uncertainty and increasing personalization.

Main Components of Virtual Try-On Systems

1. Pose Detection

Pose detection identifies human body keypoints such as shoulders, elbows, hips, and knees.

Common models used for this are MediaPipe and OpenPose.

It is used for Body alignment, Clothing positioning and Movement tracking.

2. Body Segmentation

Segmentation separates the person from the background and identifies body regions.

It is used for Accurate cloth fitting, Background removal and Better body measurements.

3. Cloth Warping Models

These models reshape clothing images to match the user's body pose and size.

It is used for Realistic cloth deformation, Proper cloth alignment and Improved fitting visualization.

4. Image Generation Models

Generative AI creates the final realistic try-on image. Stable Diffusion-based virtual try-on models

Used for Realistic clothing appearance, Texture preservation and Natural image synthesis.

5. Recommendation Systems

AI recommendation systems suggest clothes based on user data and preferences.

It suggests based on body measurements, fashion preferences, purchase history and similar users.

CHALLENGES FACED:

- Handling different body poses
- Accurate cloth fitting
- Occlusion handling
- Realistic texture generation
- Real-time processing speed
- Complex backgrounds and lighting